Project Executable Files

Date	
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for the project. All files are properly organized, named, and uploaded to the specified submission platform. The evaluator can run or deploy the solution using the files and instructions provided below.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA)	Details
1	Complete source code (all files and folders)	Yes	
2	README / Setup Guide (instructions to run the project)	Yes	
3	requirements.txt / package.json / dependency file	Yes	
4	Database schema / seed files (if applicable)	NA	
5	Environment configuration file (.env.example or similar)	NA	
6	Deployed application URL (if hosted)	Yes	https://opticrop-407a.onrender.com
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA	
8	Dockerfile / Containerization config (if applicable)	NA	
9	Test files and test results	Partial	Tested manually with multiple input value sets
10	Demo video or walkthrough (if required)	Yes	

Step 2: File / Folder Structure

Overview of the project's file and folder structure:

```
OptiCrop/
├── app.py
├── crop_model.pkl
├── scaler.pkl
├── OptiCrop.ipynb
├── requirements.txt
├── README.md
├── data/
│     └── Crop_recommendation.csv
├── templates/
     ├── home.html
     ├── about.html
     ├── findyourcrop.html
     └── result.html
```

Step 3: Deployment / Access Details

Hosted / Deployed URL	https://opticrop-407a.onrender.com
Login Credentials (Demo)	Public access
Platform / Hosting Provider	Render
Repository Link	NA
Demo Video Link	NA

Step 4: Run Instructions

Steps for the evaluator to run the project locally:

1. Install Python 3.10 or later.
2. Download or clone the project files.
3. Open the project folder in Command Prompt or Terminal.
4. Install the required dependencies:

```
pip install -r requirements.txt
```

5. Start the Flask application:

```
python app.py
```

6. Open the URL displayed in the terminal (usually http://localhost:5000) in your web browser.

7. Enter the required agricultural parameters on the Find Your Crop page and click Predict to get the crop recommendation.

Pre-requisites:

- Python 3.10 or above
- Flask
- Pandas
- NumPy
- Scikit-learn
- Pickle

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Prediction accuracy depends on the quality and correctness of user input.	Users should provide accurate soil and environmental parameters.
2	The model supports only the 22 crop classes included in the training dataset.	More crop datasets can be added in future versions to improve coverage.
3	Internet connection is required to access the hosted Render application.	Run the application locally using python app.py if internet is unavailable.